Pharmacy Medicine Stock and Billing Management System

SUBMITTED BY

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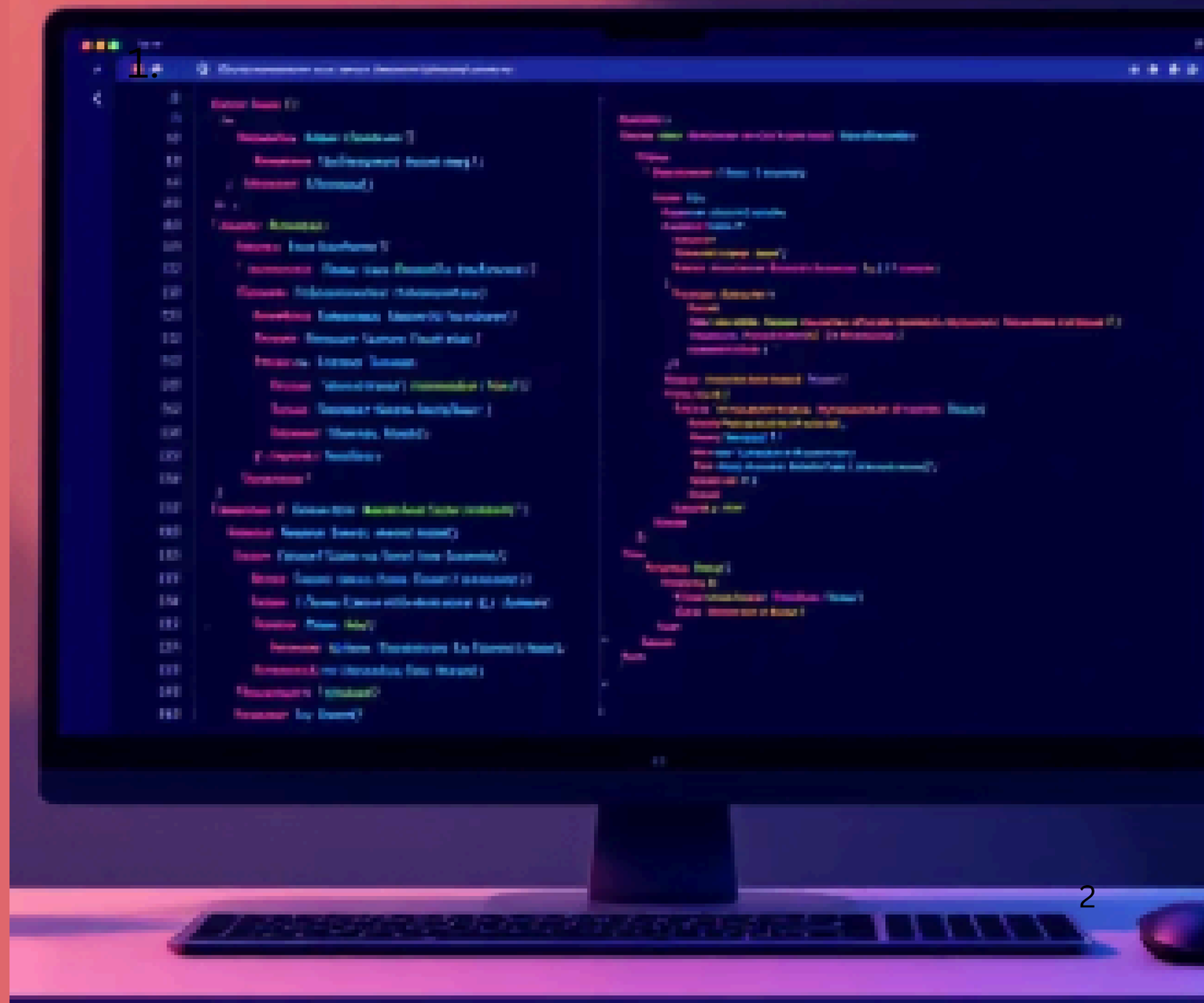
- A Java-based application automating pharmacy operations, managing medicine stock, tracking expiry dates, and generating accurate bills with GST calculation.

- Manual pharmacy operations are often time-consuming, inefficient, and prone to errors in stock updates, billing, and expiry monitoring. This project addresses those challenges by digitizing routine tasks and improving accuracy across day-to-day pharmacy management.

- The system is developed using Core Java concepts such as classes, objects, methods, exception handling, and file handling. These features help structure the application logically, manage user interactions reliably, and support secure processing of pharmacy records.



Why this System?

MANUAL ERRORS

- Handwritten registers and calculators lead to billing mistakes and stock mismanagement

TIME-CONSUMING

- Manual billing slows service, increasing customer waiting time

SAFETY RISKS

- Poor expiry tracking may result in selling expired medicines

Key Objectives

01

AUTOMATE STOCK MANAGEMENT

- Track medicine quantities and update automatically after each sale

02

ACCURATE BILLING

- Calculate totals and GST automatically, eliminating manual errors

03

EXPIRY TRACKING

- Monitor expiry dates to prevent selling expired products

04

EFFICIENT SERVICE

- Reduce waiting time and improve customer satisfaction

System Features

MEDICINE MANAGEMENT

- Add, update, and delete medicine records with name, price, quantity, and expiry date

STOCK CONTROL

- Maintain accurate stock levels and track availability in real-time

BILLING SYSTEM

- Generate structured bills with itemized details, subtotal, GST, and final amount

DATA PERSISTENCE

- Save and retrieve data using file handling for permanent storage

ERROR HANDLING

- Exception handling prevents crashes and ensures reliable operation

QUICK SEARCH

- Easily find and access medicine details when needed

System Architecture

MEDICINE MANAGEMENT

- The medicine module is built around structured classes and data containers that store each record as a complete object. It tracks key attributes such as medicine name, ID, price, quantity, and expiry date, making it easy to add, update, delete, and search records consistently.
- Medicine data is typically maintained in an in-memory collection such as a list or array during runtime, then synchronized with persistent storage when changes occur. This design keeps the record format uniform and makes validation simpler before any operation is committed.

STOCK DISPLAY

- The stock display component retrieves inventory data directly from the stored medicine records and renders it in a readable format for the user. It shows available quantities, item names, and other relevant fields so the current inventory state is always visible.
- Whenever a medicine is added, updated, or removed, the display can refresh immediately to reflect the latest data. This real-time update behavior helps prevent stale information and supports accurate inventory tracking during operations.

BILLING

- The billing step begins when the user selects medicines from the available stock list and enters the required quantities. Each selection is validated against the current inventory to ensure the requested amount does not exceed available stock and that only valid items are added to the bill.
- Selected items are stored in a temporary billing collection so they can be processed together before finalizing the transaction. This separation between inventory data and billing data helps keep the workflow organized and reduces the chance of accidental stock corruption

CALCULATION

- The calculation engine computes each line item using the formula price • quantity, then aggregates all line totals to produce the subtotal. After that, GST is applied at 5% of the subtotal to calculate the tax amount in a consistent and transparent way.
- The final amount is derived by adding the GST to the subtotal, giving the complete payable bill total. Performing these calculations separately makes the logic easier to test, debug, and reuse across different billing scenarios.

INVOICE

- Once the bill is finalized, the system generates an invoice that includes item names, quantities, individual prices, subtotal, GST, and the grand total. This document provides a clear summary of the transaction and serves as the official billing record for the customer.
- The invoice is then written to a file for permanent storage, allowing it to be retrieved or reviewed later if needed. File-based saving ensures the billing history remains available even after the program closes, supporting auditability and record keeping.

Implementation Methodology

PROBLEM ANALYSIS

- Identified manual system challenges: billing errors, stock mismanagement, expiry tracking issues

SYSTEM DESIGN

- Structured into modules with user-friendly menu-driven interface

PROGRAM DEVELOPMENT

- Implemented using Java with classes, arrays, loops, and file handling

TESTING & VALIDATION

- Tested with various inputs to ensure accuracy and reliability

How It Works

- Show Medicines
- Check Stock
- Update & Display

- Show Medicines
- Calculate Bill

Sample Output

MENU DISPLAY

- System shows available medicines with prices and quantities

BILL GENERATION

- Calculates: Item Total = Price ⋅ Quantity, GST = 5% of Subtotal, Final Bill = Subtotal + GST

STOCK UPDATE

- Automatically reduces medicine quantity after each sale

Conclusion & Impact

KEY ACHIEVEMENTS

- Reduced manual effort and calculation errors
- Improved billing speed and accuracy
- Enhanced stock tracking and expiry monitoring
- Demonstrated practical Java OOP concepts

FUTURE ENHANCEMENTS

- Can be extended with Java Swing GUI, database integration, multi-user support, and advanced reporting features.